Growing Patterns

Name _____ **Date** _____ No calculator **Version 1**

KEY WORDS pattern · term · rule · increase · decrease · input · output

A Adding and subtracting patterns ●○○ — ANSWER ON THE SHEET

Remember: Find what is added or subtracted each time. Check that it is the same between every pair of numbers.

1. Write the next two terms of each pattern.

a) 3, 7, 11, 15, ___, ___

b) 5, 11, 17, 23, ___, ___

c) 20, 17, 14, 11, ___, ___

d) 2, 11, 20, 29, ___, ___

e) 50, 43, 36, 29, ___, ___

f) 8, 19, 30, 41, ___, ___

g) 100, 85, 70, 55, ___, ___

h) 125, 150, 175, 200, ___, ___

B Decimals and fractions ●●○ — ANSWER ON THE SHEET

2. Write the next two terms of each pattern.

a) 0.5, 1, 1.5, 2, ___, ___

b) 0.2, 0.4, 0.6, 0.8, ___, ___

c) 1.5, 1.75, 2, 2.25, ___, ___

d) 3, 2.5, 2, 1.5, ___, ___

e) $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1, ___, ___

f) $\frac{1}{3}$, $\frac{2}{3}$, 1, $1\frac{1}{3}$, ___, ___

C Multiplying patterns ●●○ — ANSWER ON THE SHEET

Remember: Some patterns grow by multiplying. In 2, 4, 8, 16, … each number is double the one before.

3. Write the next two terms of each pattern.

a) 1, 2, 4, 8, ___, ___

b) 1, 3, 9, 27, ___, ___

c) 5, 10, 20, 40, ___, ___

d) 3, 30, 300, 3,000, ___, ___

e) 96, 48, 24, 12, ___, ___

f) 2, 6, 18, 54, ___, ___

D Describing a rule ●●○ — WORK IN YOUR BOOK

4. Describe the rule for each pattern in words.

a) 4, 7, 10, 13, 16

b) 30, 25, 20, 15, 10

c) 2, 4, 8, 16, 32

d) 7, 7.5, 8, 8.5, 9

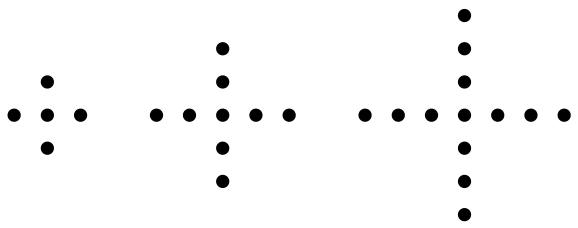
e) 1, 10, 100, 1,000, 10,000

f) 64, 32, 16, 8, 4

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

E Patterns from diagrams ●●● — WORK IN YOUR BOOK

5. Dots are drawn in the shape of a plus sign.



Pattern 1 **Pattern 2** **Pattern 3**

a) Complete the table.

Pattern	1	2	3	4	5
Number of dots					

b) How many dots are added each time?

c) How many dots are there in pattern 8?

F Predicting further terms ●●● — WORK IN YOUR BOOK

6. Answer each question.

a) The pattern 5, 8, 11, 14, … carries on in the same way. What is the 10th term?

b) The pattern 4, 10, 16, 22, … carries on in the same way. What is the 8th term?

c) The pattern 50, 46, 42, 38, … carries on in the same way. What is the 10th term?

G Tables of values from a rule ●●● — ANSWER ON THE SHEET

> **Remember:** Put each input into the rule to find its output. For the rule "multiply the input by 2, then add 1", the input 3 gives $3 \times 2 + 1 = 7$.

7. Use the rule to complete each table.

a) Rule: add 5

Input	1	2	3	4	5
Output					

b) Rule: multiply the input by 3

Input	1	2	3	4	5
Output					

c) Rule: multiply the input by 2, then add 1

Input	1	2	3	4	5
Output					

d) Rule: multiply the input by 3, then subtract 1

Input	2	4	6	8	10
Output					

★ Extension — FOR EARLY FINISHERS

8. The pattern 3, 6, 12, 24, … is multiplied by 2 each time. What is the first term that is larger than 100?

9. Look at the pattern 1, 4, 9, 16, … Write the next two terms. What is added each time?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Cell Tuition
1-1 PRIVATE TUTORS

YEAR 6 · ALGEBRA

Growing Patterns

Name _____ **Date** _____ No calculator Version 2

KEY WORDS pattern · term · rule · increase · decrease · input · output

A Adding and subtracting patterns ●○○

ANSWER ON THE SHEET

Remember: Find what is added or subtracted each time. Check that it is the same between every pair of numbers.

1. Write the next two terms of each pattern.

a) 2, 7, 12, 17, ___, ___

b) 4, 11, 18, 25, ___, ___

c) 25, 21, 17, 13, ___, ___

d) 3, 11, 19, 27, ___, ___

e) 60, 51, 42, 33, ___, ___

f) 7, 19, 31, 43, ___, ___

g) 90, 75, 60, 45, ___, ___

h) 150, 175, 200, 225, ___, ___

B Decimals and fractions ●●○

ANSWER ON THE SHEET

2. Write the next two terms of each pattern.

a) 1.5, 2, 2.5, 3, ___, ___

b) 0.3, 0.6, 0.9, 1.2, ___, ___

c) 2.5, 2.75, 3, 3.25, ___, ___

d) 4, 3.5, 3, 2.5, ___, ___

e) $\frac{1}{2}$, 1, $1\frac{1}{2}$, 2, ___, ___

f) $\frac{1}{5}$, $\frac{2}{5}$, $\frac{3}{5}$, $\frac{4}{5}$, ___, ___

C Multiplying patterns ●●○

ANSWER ON THE SHEET

Remember: Some patterns grow by multiplying. In 2, 4, 8, 16, … each number is double the one before.

3. Write the next two terms of each pattern.

a) 3, 6, 12, 24, ___, ___

b) 2, 6, 18, 54, ___, ___

c) 5, 10, 20, 40, ___, ___

d) 4, 40, 400, 4,000, ___, ___

e) 160, 80, 40, 20, ___, ___

f) 1, 4, 16, 64, ___, ___

D Describing a rule ●●○

WORK IN YOUR BOOK

4. Describe the rule for each pattern in words.

a) 5, 9, 13, 17, 21

b) 40, 34, 28, 22, 16

c) 1, 3, 9, 27, 81

d) 3, 3.5, 4, 4.5, 5

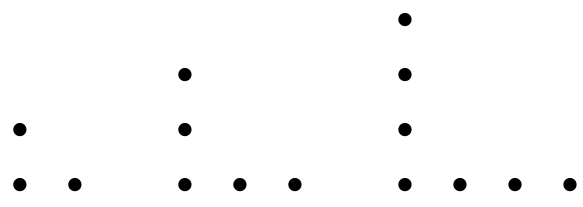
e) 2, 20, 200, 2,000, 20,000

f) 80, 40, 20, 10, 5

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

E Patterns from diagrams ●●● — WORK IN YOUR BOOK

5. Dots are drawn in the shape of a letter L.



Pattern 1 **Pattern 2** **Pattern 3**

a) Complete the table.

Pattern	1	2	3	4	5
Number of dots					

b) How many dots are added each time?

c) How many dots are there in pattern 8?

F Predicting further terms ●●● — WORK IN YOUR BOOK

6. Answer each question.

a) The pattern 6, 10, 14, 18, … carries on in the same way. What is the 10th term?

b) The pattern 3, 10, 17, 24, … carries on in the same way. What is the 8th term?

c) The pattern 60, 55, 50, 45, … carries on in the same way. What is the 10th term?

G Tables of values from a rule ●●● — ANSWER ON THE SHEET

> **Remember:** Put each input into the rule to find its output. For the rule "multiply the input by 2, then add 1", the input 3 gives $3 \times 2 + 1 = 7$.

7. Use the rule to complete each table.

a) Rule: add 7

Input	1	2	3	4	5
Output					

b) Rule: multiply the input by 4

Input	1	2	3	4	5
Output					

c) Rule: multiply the input by 2, then add 3

Input	1	2	3	4	5
Output					

d) Rule: multiply the input by 4, then subtract 1

Input	2	4	6	8	10
Output					

★ Extension — FOR EARLY FINISHERS

8. The pattern 2, 6, 18, 54, … is multiplied by 3 each time. What is the first term that is larger than 200?

9. Look at the pattern 1, 4, 9, 16, … Write the next two terms. What is added each time?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.